Pathology: Nephritic Syndrome

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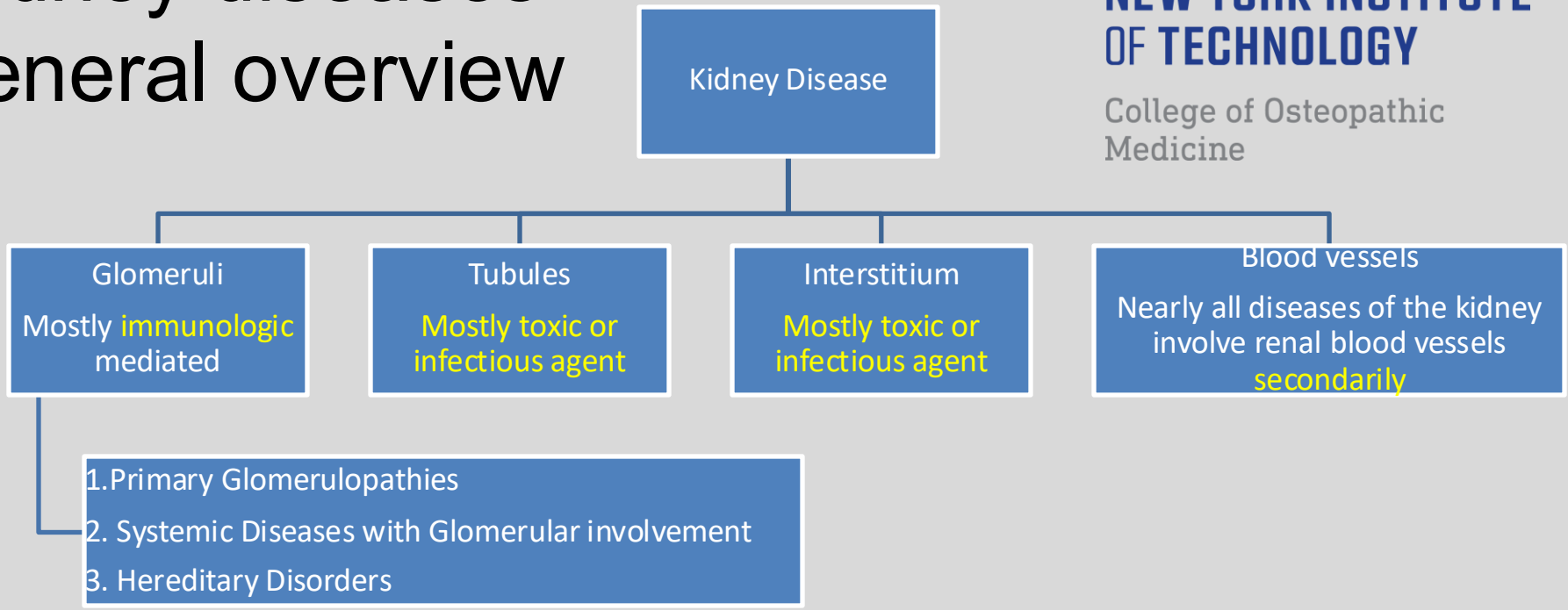
Session Objectives

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1. Following a brief review of renal pathophysiology, interpret the pathogenesis, pathophysiology, pathology and clinicopathology of **IgA nephropathy**
2. Identify the pathogenesis, pathophysiology, pathology and clinicopathology of poststreptococcal glomerulonephritis (**PSGN**).
3. Contrast the pathogenesis, pathophysiology, pathology and clinicopathology of etiologic variants of RPGN (e.g. **Goodpasture** syndrome), and compare it with pauci-immune glomerulonephritis (**Granulomatosis with polyangiitis**).
4. Distinguish the pathogenesis, pathophysiology, pathology and clinicopathology of the forms of **lupus nephritis**.
5. Analyze the relationships between **hereditary** forms of nephritic syndrome with regard to the pathogenesis, pathophysiology, pathology and clinicopathology of **Alport syndrome** & **thin basement membrane disease**.

Kidney diseases general overview

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Nephritic Syndrome

Inflammatory lesions of **glomeruli** injures capillary walls

Hematuria
Red cells / red cell casts in urine

Proteinuria
Sub-nephrotic range

↓GFR

Fluid Retention

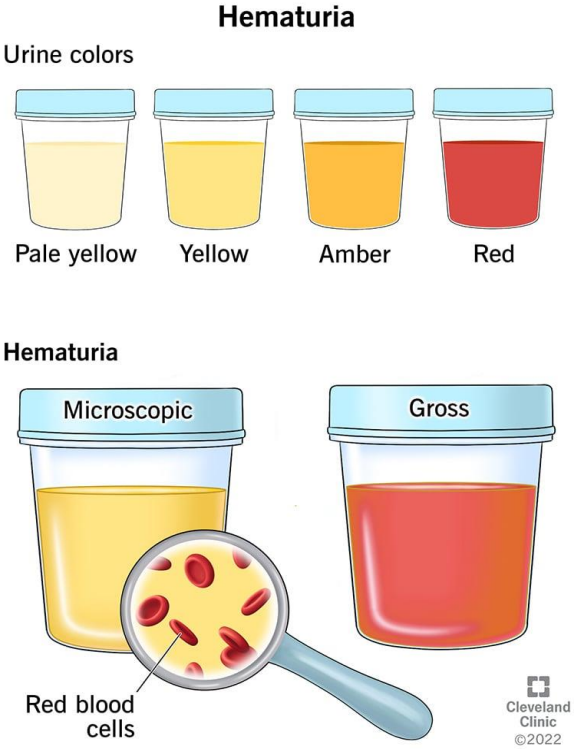
Edema

Azotemia
↑BUN/Cr

Hypertension

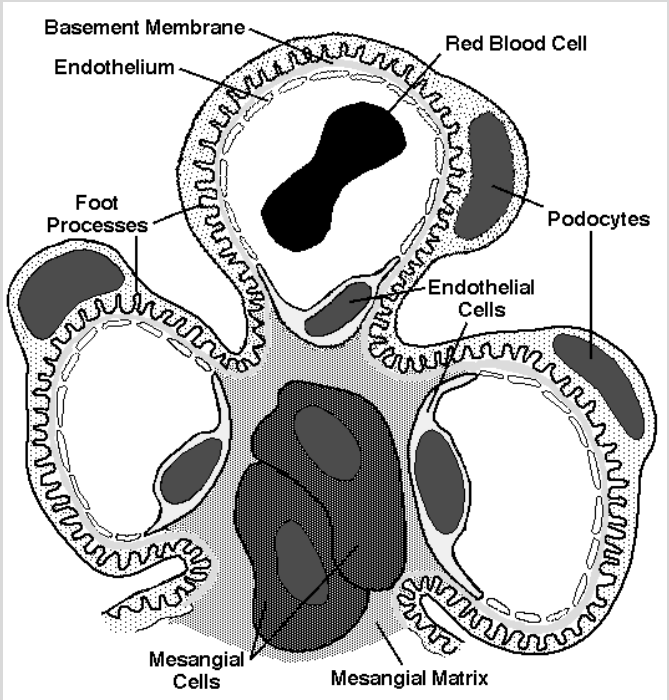
Oliguria
↓Urine output

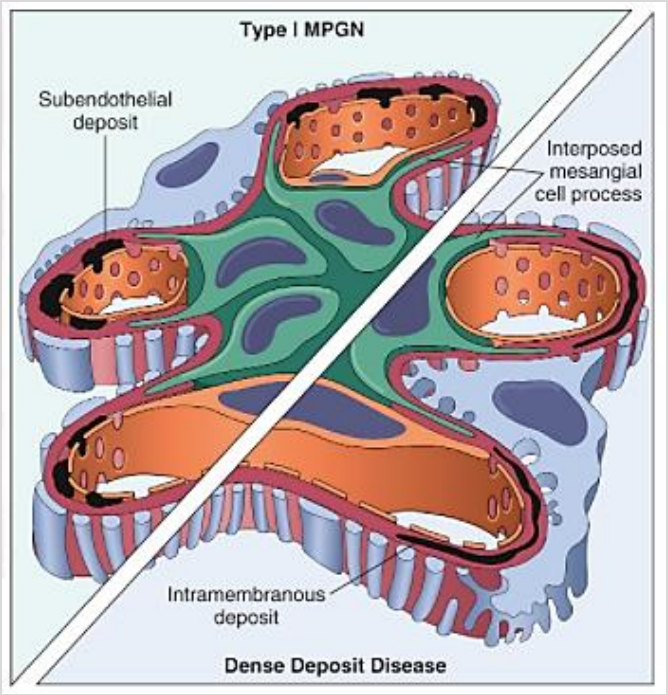
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Review of the glomerular filtration barrier

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Shown is a MPGN diagram w/ deposits, but also shows filtration barrier layers well, including the basis or the name of deposits

IgA Nephropathy (Berger Disease)

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- **Most common** primary glomerulonephritis worldwide

- Major cause in older **children** & **younger** adults
- First-seventh decade of life; peak 20-30 years
- **M**>F

- Clinically, has macroscopic & microscopic **hematuria** w/ or w.o. **proteinuria**

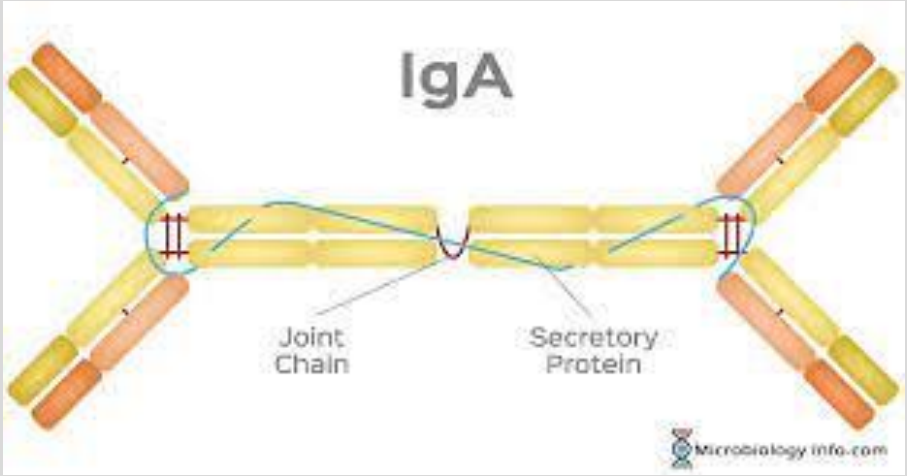
- 80% of patients show renal survival in 10 years
- 25-50% progress to end stage renal disease > 20 years of disease activity

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Berger Disease pathogenesis

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- Pathogenesis: Increased mucosal **IgA** synthesis in response to mucosal tract **infection (respiratory** or **GI tract)**
  - ↑ **serum** levels of abnormally/poorly galactosylated (galactose-deficient IgA1, **GD-IgA**) **IgA**
  - **Anti-GD-IgA** antibody formation
- **IgA immune complexes** deposit in mesangium→ activate alternative **complement** pathway & initiate glomerular injury



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Berger Disease pathology

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- Light/Electron microscopy(**LM**/**EM**): Mesangial **IgA** immune complexes (**ICs**) deposits & **mesangial expansion**
  - Mesangial cell proliferation
  - IgA /anti-GD-IgA **complex** deposition in mesangium
  - Mesangial deposition of **C3** & **properdin**
  - Cytokine release by mesangial cells leading to podocyte injury

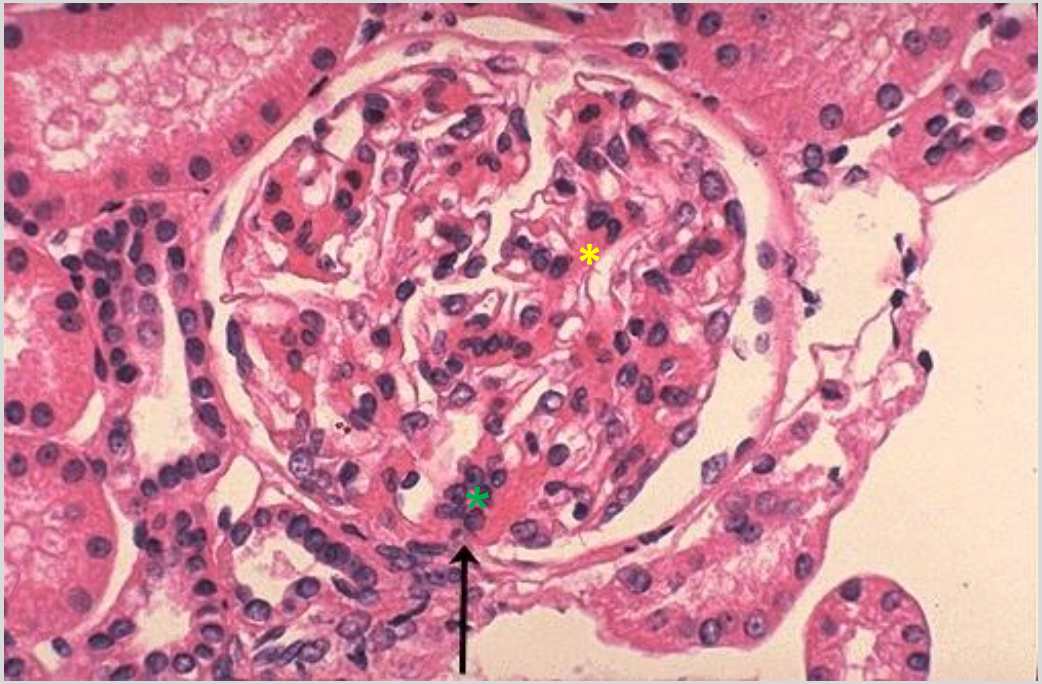
- Immunofluorescent microscopy (**IF**) pattern **granular**

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Berger Disease LM

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Tissue Periodic Acid Schiff stain, **PAS**: ↑ mesangial protein* deposition & mesangial cell*(arrow) proliferation

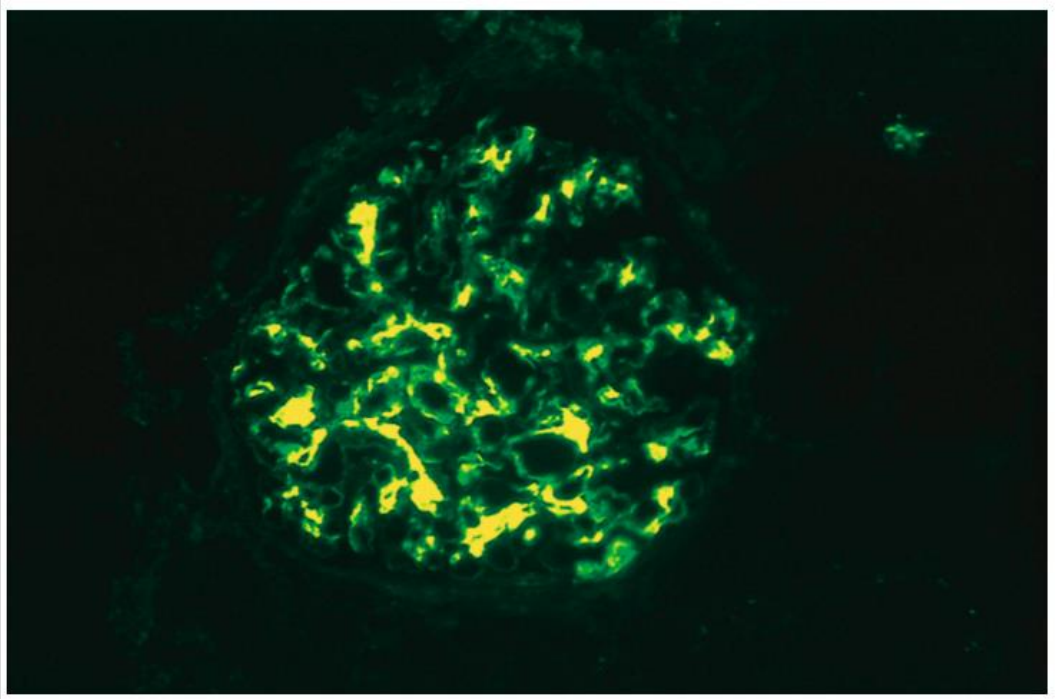


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Mesangial deposits

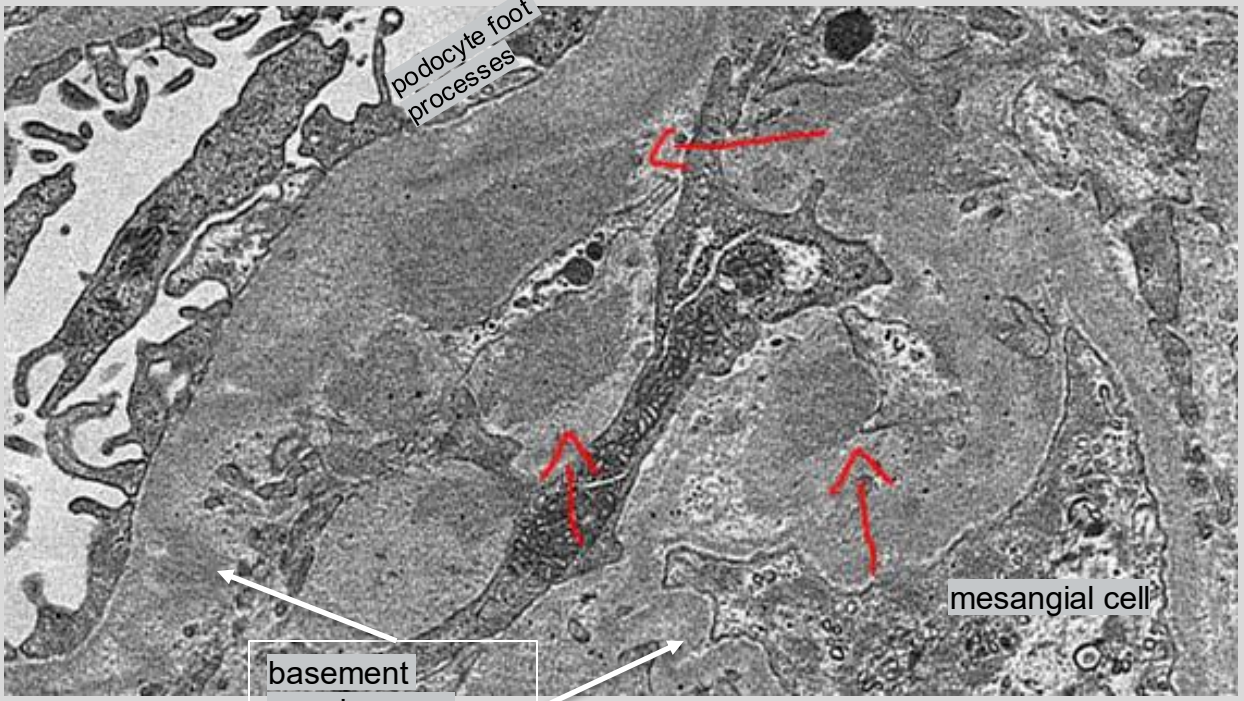


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Mesangial deposits, red arrows



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Berger Disease clinical presentation

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- **Recurrent hematuria**, mild **proteinuria**, ↑ **serum IgA**
- Clinical predictors for disease progression
  - High serum **creatinine** / ↓ GFR
  - **Hypertension** > 140/90 mmHg
  - **Proteinuria** for over 6 months
- Associated w/ **IgA vasculitis**, formerly Henoch- Schönlein purpura**,** intestinal diseases (like **Celiac Disease**) & **liver disease**
- **Post-infection** of mucosal tract (i.e., within a few days)
  - Example presentation: 12-year-old boy brought in by his mother because of swollen legs and “dark urine.” ROS noted symptoms consistent with URI a few days ago. Vital signs reveal an elevated BP. UA reveals numerous RBC’s and RBC casts, as well as some protein and WBC’s

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Poststreptococcal Glomerulonephritis (PSGN)

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- Usually seen in **children**, 2-6 years old & has a **good prognosis**
  - Infection related glomerulonephritis in adults has a poorer outcome

  - Pathogenesis: Commonly follows **group A streptococci** infection (1-4 weeks later)
    - Sources of infection: **Skin** (impetigo, erysipelas), **pharynx**
    - Nephritogenic strains have cell surface M protein

    - Immune **complexes** leads to tissue injury primarily by complement activation (serum **complement** ↓)
    - Serology tests antistreptococcal antibodies: Antistreptolysin O **(ASO)**

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PSGN histopathology

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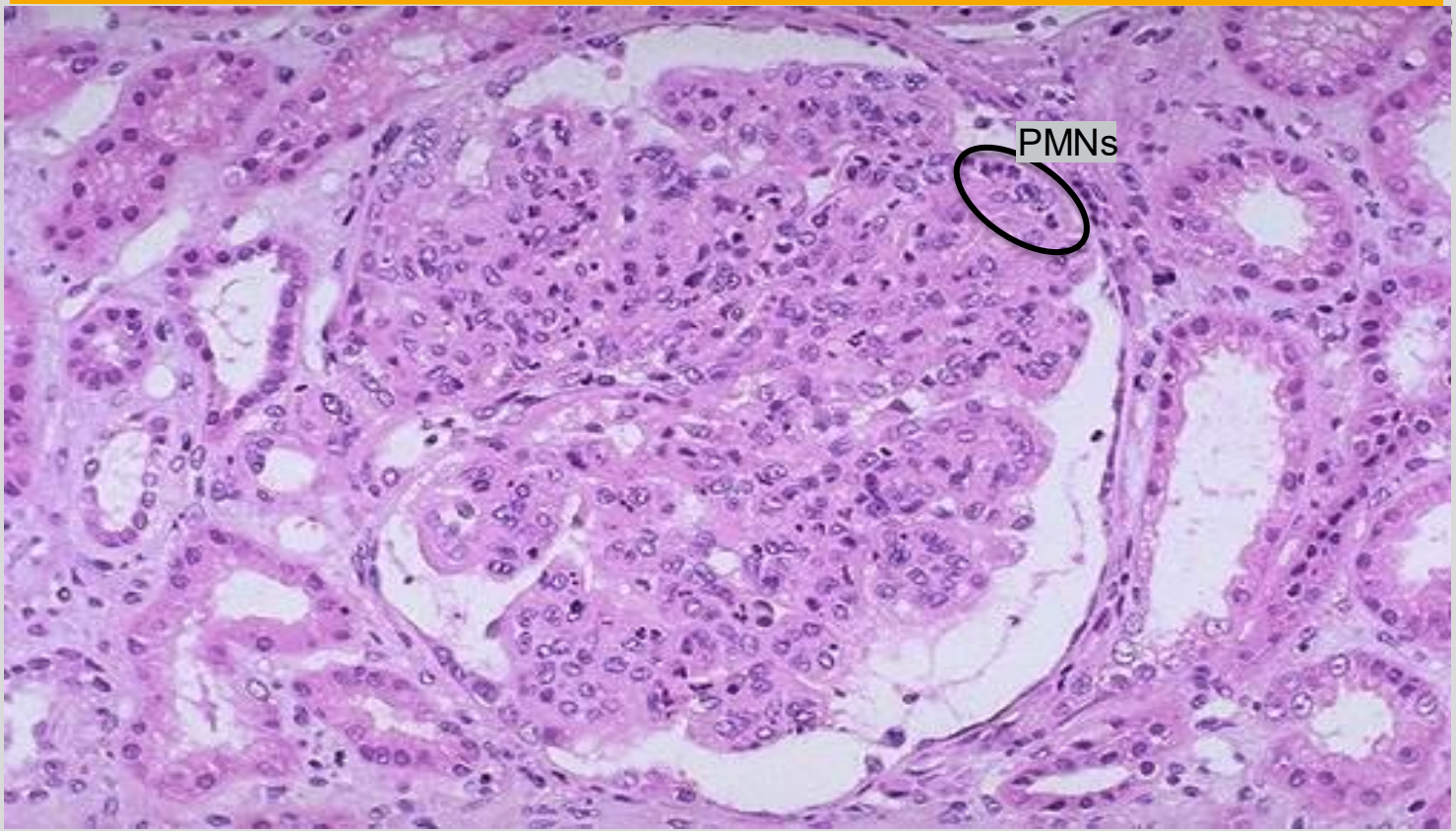
- LM: **Diffuse hypercellular proliferative pattern** with neutrophil (**PMNs**) & monocyte infiltration, proliferation of endothelial & mesangial cells, endocapillary hypercellularity & **crescents** if severe
- IF: Scattered granular deposits, small, irregular, **coarse granular** deposits along glomerular capillary walls w/ scattered mesangial deposits "**starry sky**"/"**lumpy, bumpy**", **IgG, C3**
- EM: **Hump-type subepithelial** (the podocytes) deposits, occasional mesangial deposits & rare subendothelial deposits w/o significant glomerular basement membrane reaction, electron-dense deposits (humps)

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PSGN LM

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H&E: Very cellular glomerulus

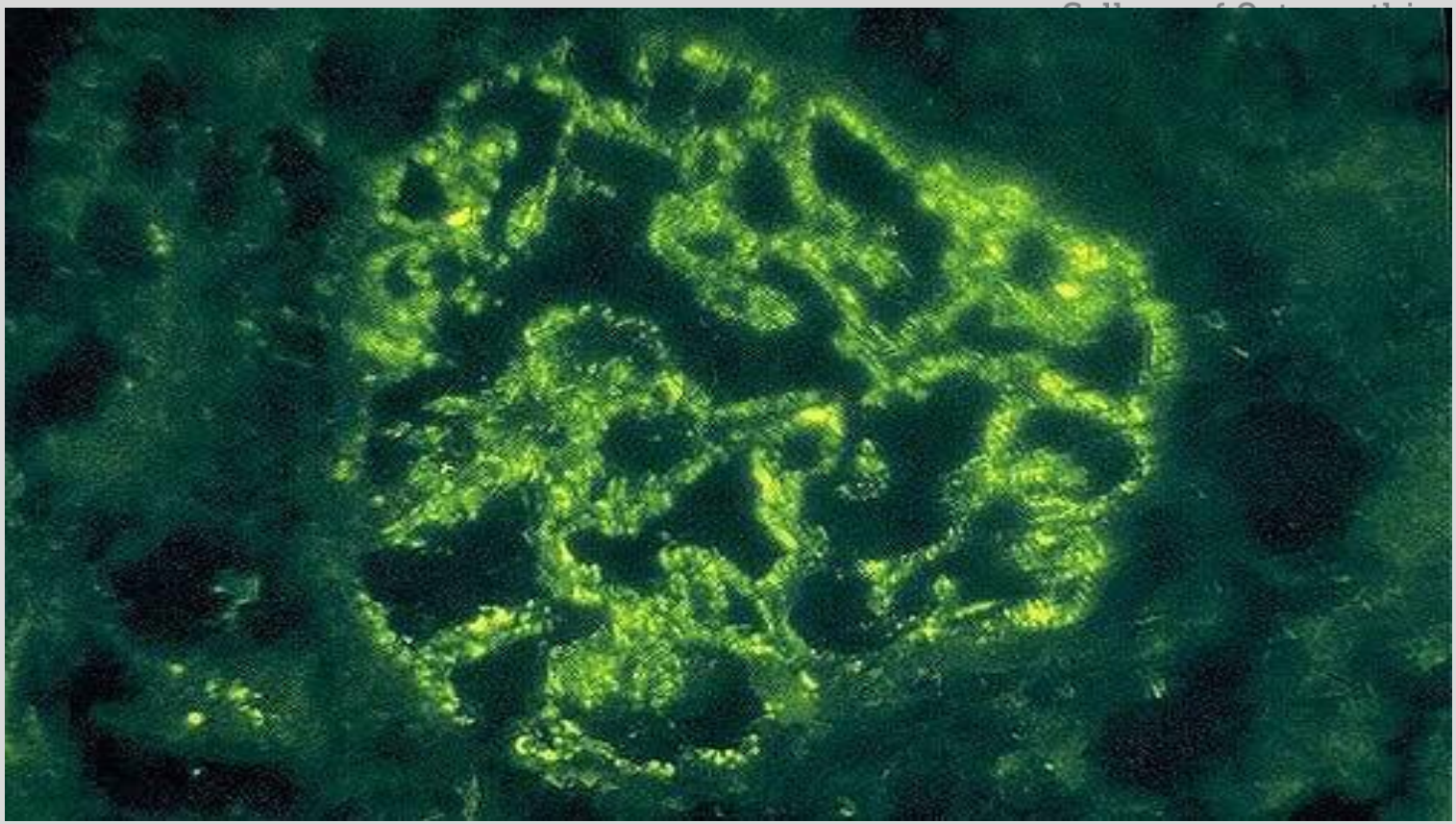


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PSGN IF

Coarse granular deposits along glomerular capillary walls w/ scattered mesangial deposits, “starry sky”

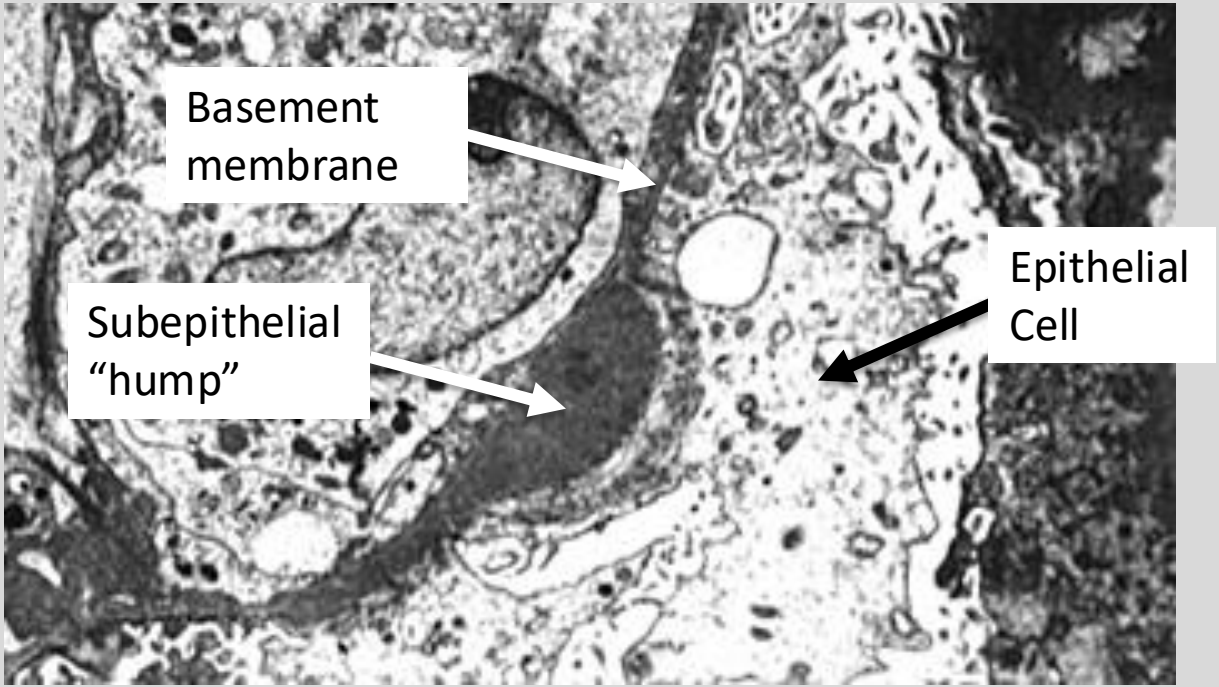
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PSGN clinical presentation & management

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- Dark **urine** (brown, tea or cola colored) is often 1st clinical manifestation
  - **Hematuria**,1-4 weeks following GAS infection
- Signs & symptoms (Sx/Sx): Periorbital edema, hypertension sometimes, generalized **edema** & other features of circulatory congestion, such as dyspnea possibly
- Nonspecific Sx/Sx include general malaise, weakness & anorexia possible

- Labs: ↓ **C3** w/ **normal C4,** positive **ASO**, Impetigo → anti-hyaluronidase antibodies

- **Excellent prognosis**, especially in children, with 95% completely recovering with conservative therapy within 6-8 weeks
  - ~50% of the adult patients continue to have reduced renal function, hypertension or persistent proteinuria
  - **Some** cases develop into rapidly progressive glomerulonephritis (**RPGN**) **pattern** of injury
- Self limiting, usually requires symptomatic treatment only
  - Main aim is to control the complications of volume overload such as hypertension and edema, which are prominent during the acute phase

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PSGN clinical example

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10-year-old presents to clinic because mother reports "**dark-red colored urine**" and states the child's **urine output** has begun to **decrease**

ROS: (+) **Sore throat 2 weeks ago**
Vital signs reveal **BP 140/90**, **Temp 101**
UA: (+) **RBC Casts**, (+) **WBC's**, (+) **Protein**
PE: (+) Swollen **Puffy Eyes**
Labs: (+) **elevated BUN/Cr** (because of decreased GFR)

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Rapidly Progressive Glomerulonephritis (RPGN)

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- A **Pattern of Injury** with **rapid deterioration** of renal function in weeks to months following onset of a nephritic syndrome, causes include the following:
  - Anti-Glomerular BM GN (**Goodpasture disease**): Check **Anti-GBM** levels
    - **IF** pattern is **Linear**
  - **Pauci-Immune GN:** Check **ANCA's**, systemic vasculitides (e.g. microscopic polyangiitis)
    - **IF** pattern**: Negative**
  - **Immune Complex GN:** Check **compliment**
    - Causes include PSGN, lupus (diffuse proliferative), IgA nephropathy
    - **IF** pattern: **Granular**

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RPGN histopathology

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- LM: **Crescent** formation **within Bowman's** space
  - Composed of **fibrin,** parietal **epithelial** cells**,** monocytes & **macrophages**
  - If crescents obliterate Bowman's space: GFR of that glomerulus → 0

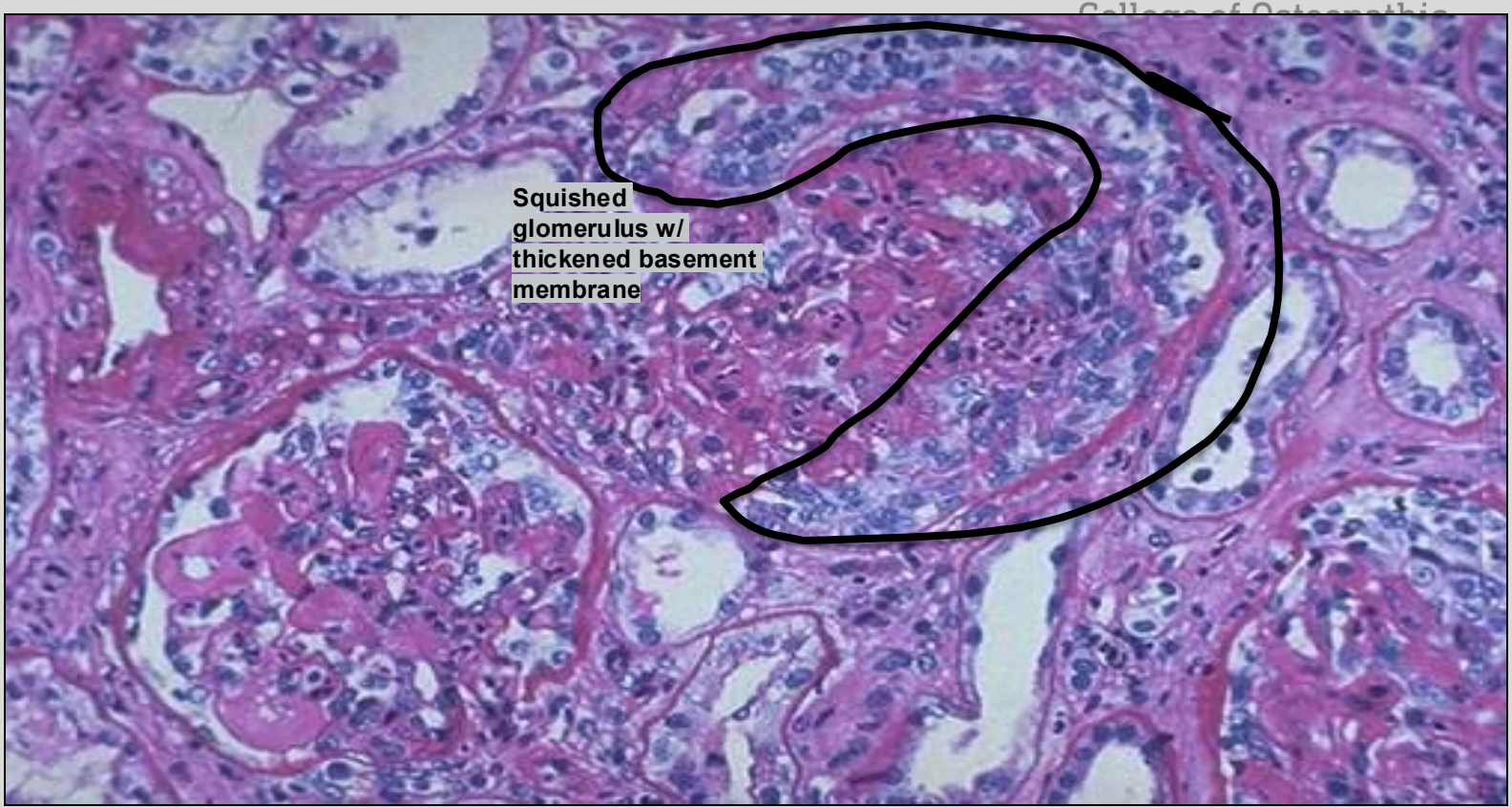
- EM: characteristic **wrinkling** of GBM w/ **focal disruptions** in its continuity

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RPGN LM - crescent

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H&E

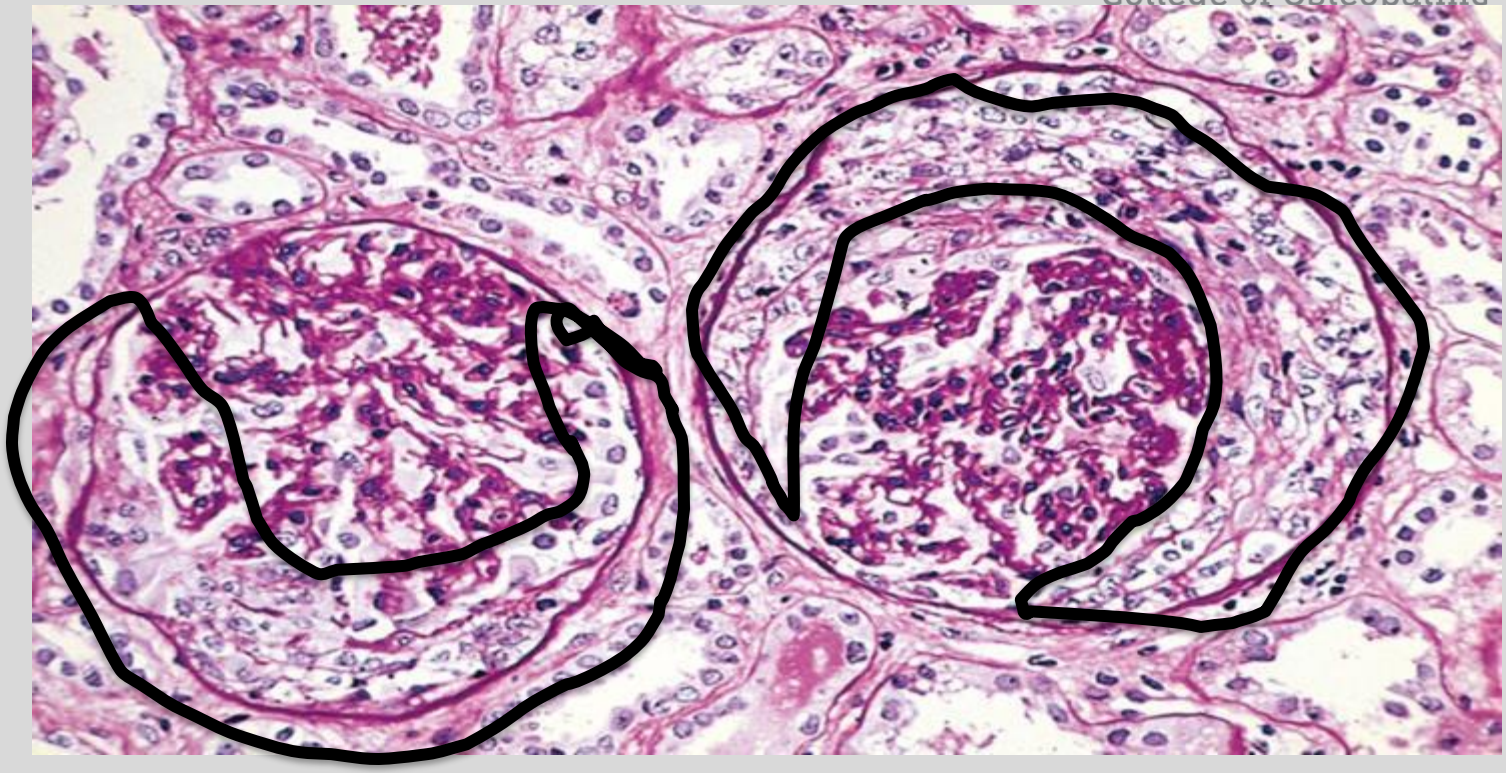


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RPGN LM crescents

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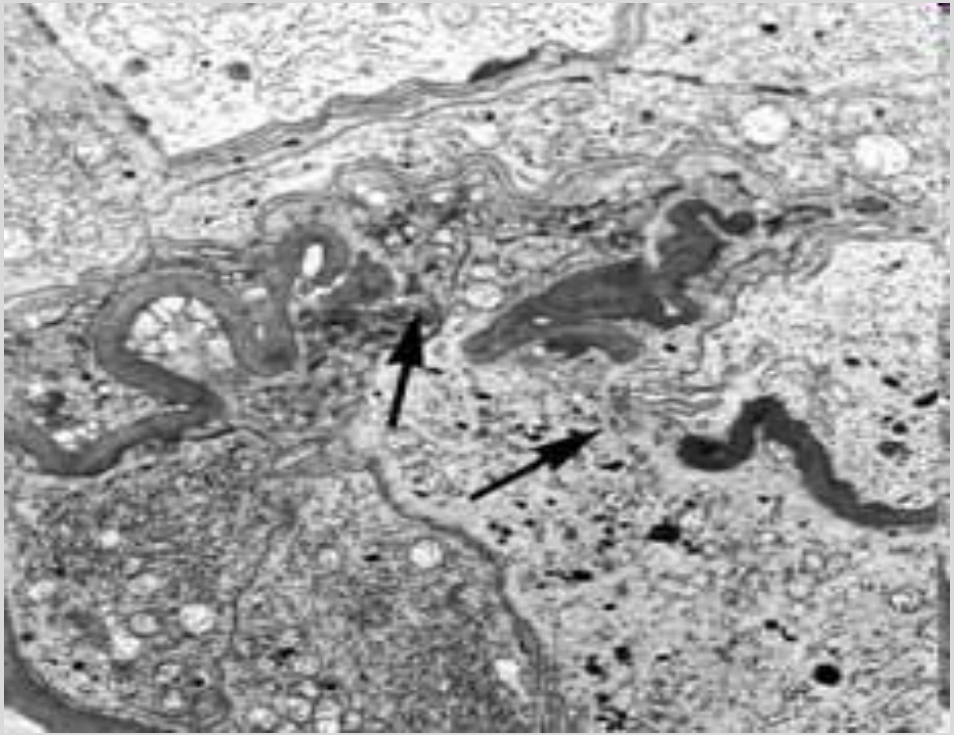


Robbins 6th Ed

RPGN EM

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Wrinkling of GBM w/focal **disruptions** in its continuity



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RPGN clinical presentation example

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38-year-old man presents **5 weeks after a sore throat**

ROS: (+) **Cola colored urine**
UA reveals (+) **Gross dysmorphic RBC's**, (+) **RBC Casts**, (+) WBC

The worrisome chemistry, Cr 3.4 Bun 66, values prompt a kidney biopsy:
Kidney Biopsy reveals (+) **Subepithelial humps**, (+) **Crescents**

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Goodpasture Syndrome (a type of RPGN) pathogenesis

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- AKA anti-glomerular basement membrane disease primarily effects **men** in their late **20's**
- Pathogenesis: **IgG** antibody (non-collagen domain (NC1) of α3 **type IV collagen** (col 4 A3)) **anti-BM antibodies** in **glomerular** & **pulmonary** capillaries
  - Antibody directed destruction, Type II hypersensitivity

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Goodpasture Syndrome Pathology

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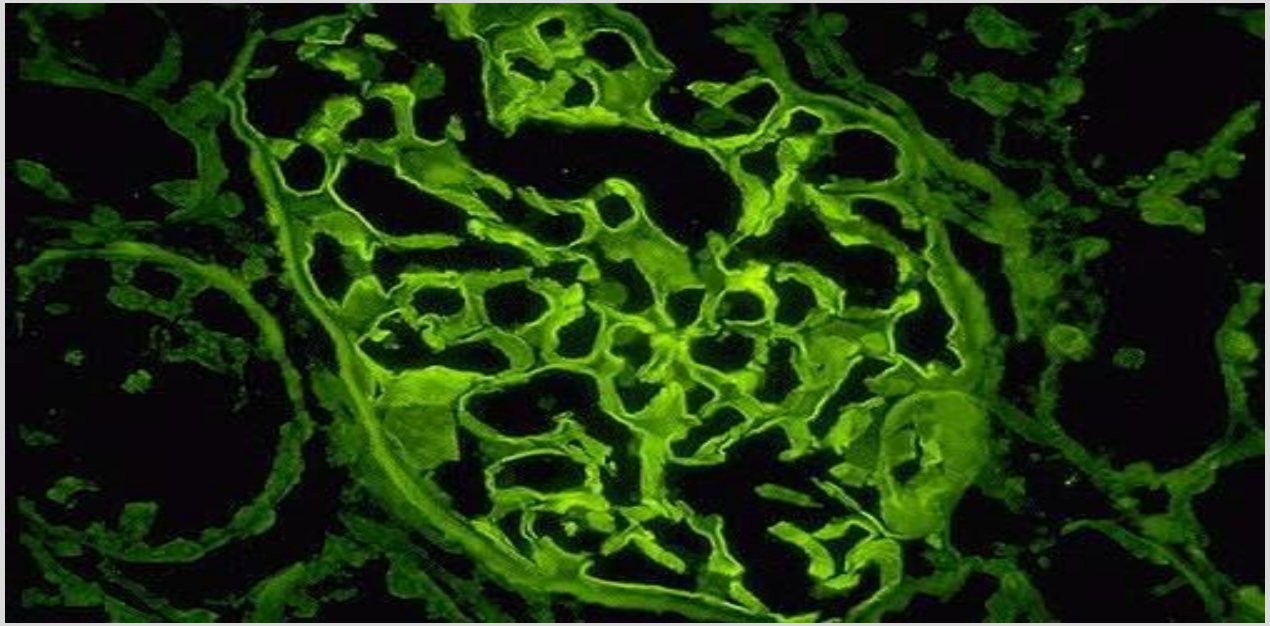
- LM: Acute, often **crescentic** glomerulonephritis
- IF: **Linear** IgG, C3
- EM: No electron dense deposits, **GBM disruptions**

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Goodpasture Syndrome IF

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Linear pattern of immunofluorescence



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Goodpasture Syndrome clinical presentation & management example

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- Clinical: Begins with **hemoptysis** & **dyspnea** & ends w/ renal failure
- Tx**:** Exchange **plasmapheresis**

- Example: **28**-year-old male presents after 3 weeks of worsening **hemoptysis**

- ROS reveals chest pain, (-) night sweats, (-) fever
- A **CXR** reveals diffuse lung **consolidations**
- UA reveals (+) **dysmorphic RBC's** & (+) RBC **casts**

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Granulomatosis with polyangiitis

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- **GPA**, formerly Wegener granulomatosis
- Rare **autoimmune** disease w/ inflammation & damage to small blood vessels
- Usually affects childhood-middle **age** groups
- Pathogenesis: Neutrophil degranulation leads to the following
    - Necrotizing med-sized & small vessel vasculitis involving
        - Upper **respiratory** tract (nose, sinuses)
        - **Lungs**
        - **Renal** vessels
    - **Leukocytoclastic** reaction
        - Neutrophils, blood vessels, necrosis
    - **Pauci-immune** vasculitis
        - Lack of significant immune cells & no deposits found on biopsy

Major vasculitides markers

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- Some vasculitides have circulating antibodies that react with anti-neutrophil cytoplasmic antigens (**ANCA**), a diagnostic marker

- **c-ANCA** (c-cytoplasmic) & **p-ANCA** (p-perinuclear)
  - Now referred to as **PR3-ANCA** (anti-proteinase-3) & **MPO-ANCA** (anti-myeloperoxidase), respectively

- Granulomatosis with polyangiitis, **GPA** (**c-ANCA**)→ pauci-immune
- **Microscopic** polyangiitis (**p-ANCA**)→ pauci-immune

GPA clinical findings

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- **Saddle nose** deformity
  - Perforation of nasal **septum/ulcer**
- Recurrent sinusitis
- **Hemoptysis**
- Pulmonary infarction
- Pneumonia

- Laboratory:
  - **Pauci-immune, crescentic** glomerulonephritis seen on **LM**
  - PR3-ANCA (**c-ANCA**)



Nasal involvement in GPA:
Vasculitis & epistaxis

From James WD eta l: Cutaneous vascular diseases. In: James WD et al, eds: Andrews' Diseases of the Skin: Clinical Atlas. Elsevier; 2018:561-81, Figure 35.80

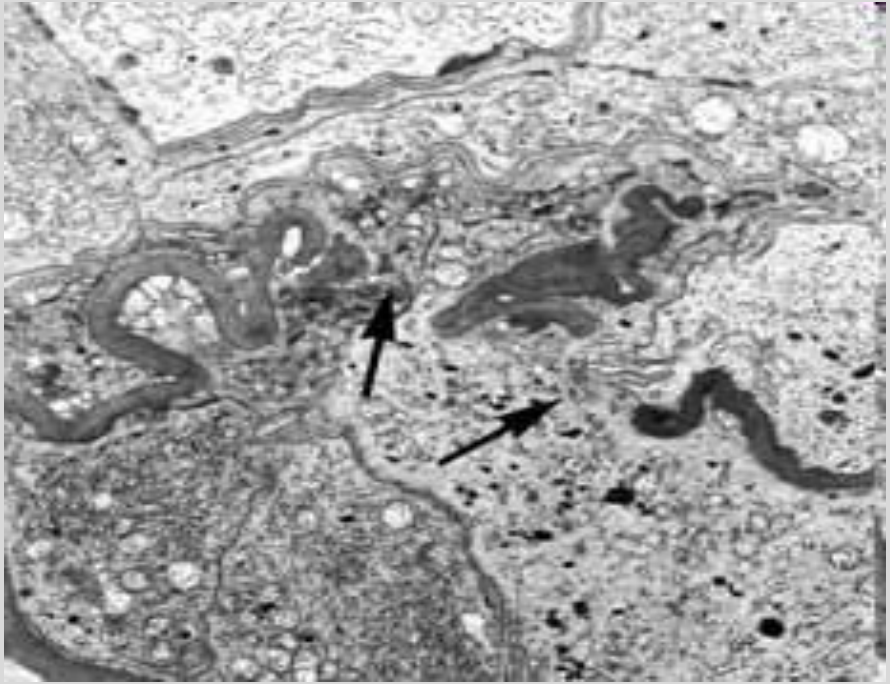
Saddle nose deformity due to bone & cartilage damage in GPA



From James WD eta l: Cutaneous vascular diseases. In: James WD et al, eds: Andrews' Diseases of the Skin: Clinical Atlas. Elsevier; 2018:561-81, Figure 35.80

GPA renal tissue EM: Fragmented GBM

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Characteristic **wrinkling** of glomerular basement membrane w/ focal **disruptions** (arrows)

Systemic Lupus Erythematosus (SLE)-Lupus **Nephritis**

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- Inflammation of the kidneys caused by SLE - a type of **glomerulonephritis** in which the glomeruli become inflamed secondary to SLE (**autoimmune**)
- 75% of SLE patients will develop kidney involvement, usually **3-6 months after SLE dx**
- 10-30% of people w/ lupus nephritis progress to kidney failure requiring **dialysis**
- 5 year mortality rate of lupus nephritis being 5-25%

- Classified into 6 "**Classes**" by histopathology:
  - Class I: **Minimal Mesangial** Nephritis
  - Class **II**: **Mesangial** Proliferative
  - Class **III**: **Focal** Proliferative Nephritis
  - Class **IV**: **Diffuse** Proliferative Nephritis
  - Class V: **Membranous** GN: **Nephrotic** syndrome
  - Class VI: Advanced **Sclerosing** Lupus Nephritis

} **Proliferative** forms are associated w/ a ↑ risk of progression → **end stage** kidney disease

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Lupus Nephritis pathogenesis & labs

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- SLE: Chronic autoimmune disease which affects **multiple** organ systems
  - Tissue damage by **autoantibodies** (due to **anti-dsDNA antibodies-**specific for SLE)
  - Anti-dsDNA is a specific type of ANA antibody found in ~ 30% with SLE

- Laboratory : Immune complex (**IC**) deposition in renal structures
  - Serum antinuclear antibody (**ANA**) test - rim pattern, ANA is highly sensitive for autoimmune diseases**,** but may have false-positive results (**non-specific** test)
  - **DNA-anti-dsDNA** ICs activate classical **complement** pathway
    - Labs show ↓ **serum C3, C4**
  - **Full-house** staining of glomeruli in renal pathology is highly suggestive of lupus nephritis
    - Full House IF (IgG, IgM, IgA, C3, C1q +) **-** if all five major immunofluorescent stains (IgA, IgG, IgM, C3, and C1Q) are positive, then it is considered as full-house immunofluorescence

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Lupus nephritis dx histopathology

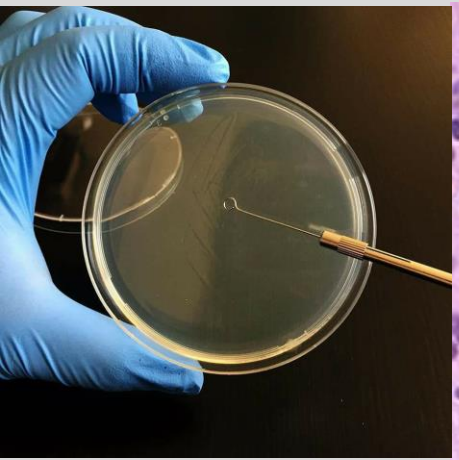
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- Depends on nephritis **class**:
  - LM: Class 1: **Normal** histology
  - Class **2**: **Mesangial** hypercellularity
  - Class **3&4**: **"wire looping"**, mesangial & endocapillary **hypercellularity**, **crescents**
  - Class 5: **GBM** thickening
  - Class 6: 'Global **sclerosis**'
  - IF: **Granular** pattern
  - EM: Class1: **Mesangial** electron dense deposits
  - Class **2**: Mesangial electron dense deposits
  - Class **3&4**: **Mesangial** & **subendothelial** electron dense deposits
  - Class 5: **Subepithelial** electron dense deposits
  - Class 6: **May not see** due to advanced disease

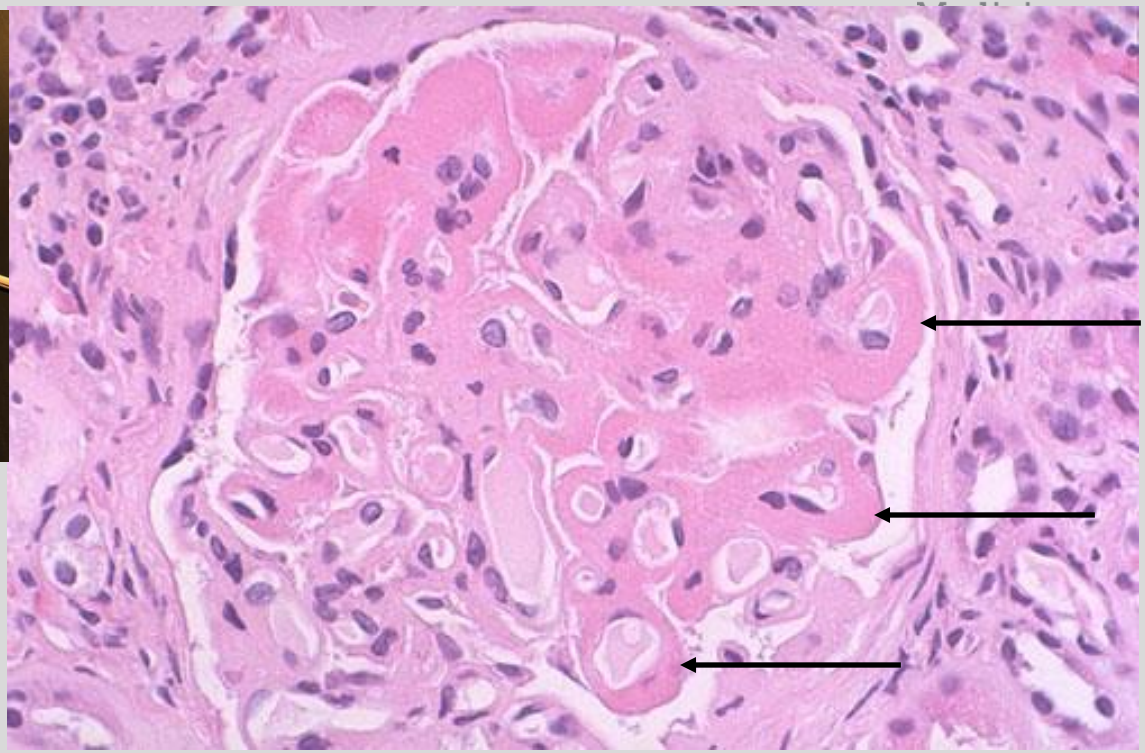
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Lupus nephritis LM

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Inoculating wire **loop** used on agar plate in microbiology lab



H&E stain

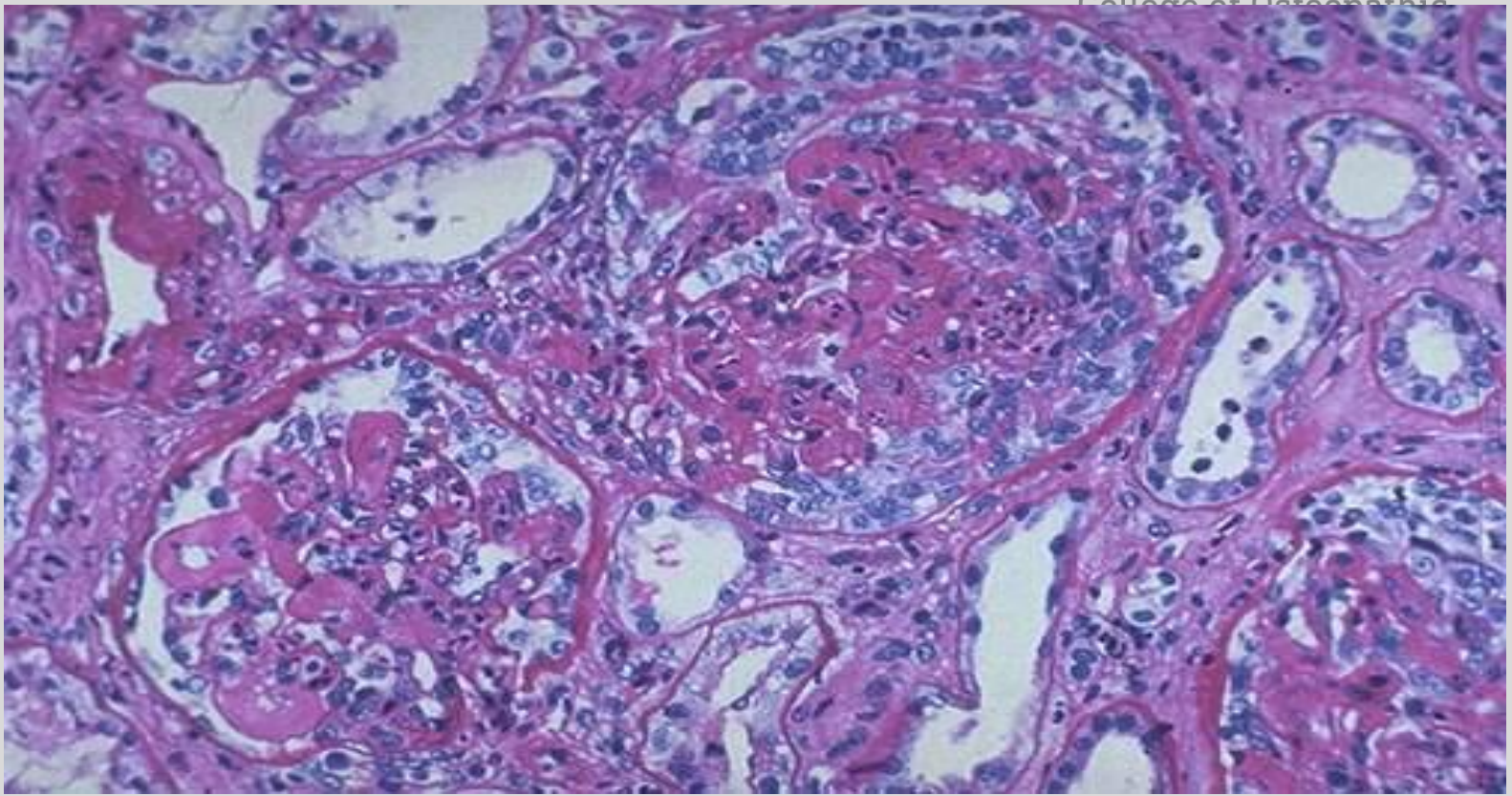
Wire-looping, very rounded & prominent **membranous** pattern of basement membranes

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Lupus nephritis LM - diffuse **proliferative** pattern

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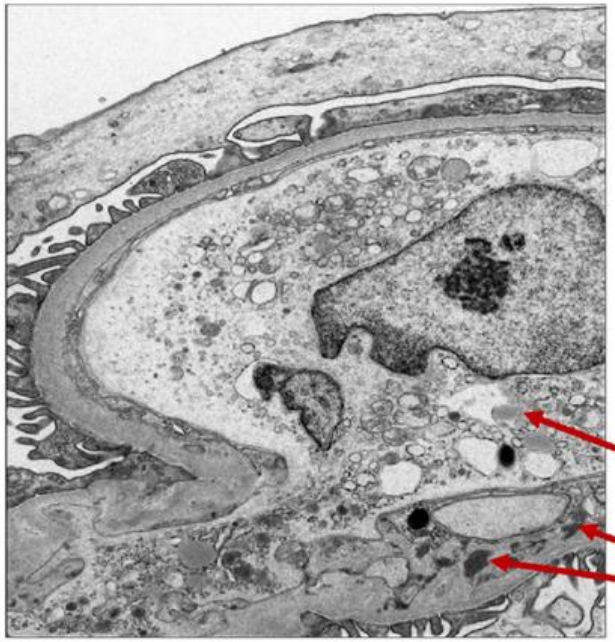
H&E stain shows **very cellular** glomeruli



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Lupus nephritis EM

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Diffuse Proliferative Glomerulonephritis with Monoclonal IgG1 Lambda Immune Deposits

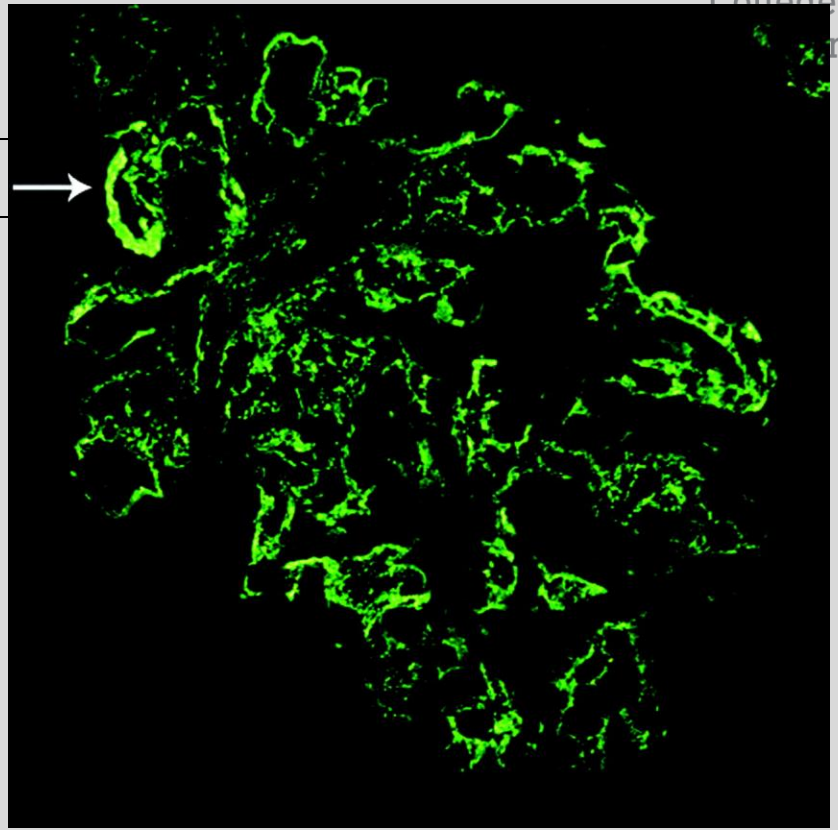
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Lupus nephritis IF

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Granular pattern



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SLE-Lupus nephritis - clinical findings

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- General Sx/Sx:
  - ~90% of people living with lupus are **women**
  - Most develop lupus between ages of **15-44 years**
  - May present w/
    - Fever
    - Fatigue
    - **Edema**
    - High blood pressure
    - Joint **pain**
    - Muscle pain
    - Malar **rash**
    - Foamy urine (**proteinuria**)

- Example presentation
  - **32**-year-old **woman** develops **rash** on her face, arthralgias, fatigue & **hematuria**
  - UA reveals gross **dysmorphic** RBC's & **RBC Casts**
  - Chemistry shows: ↑ **Cr** 2 & BUN **40** (normal 0.7-1.2 mg/dL & 6-24 mg/dL**)**
  - Kidney Biopsy reveals (+) **Wireloop** lesions, (+) **Crescents**, (+) Mesangial & Endocapillary **hypercellularity**

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Alport Syndrome

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- AKA **Hereditary** Nephritis
- Genetic disorder characterized by **glomerulonephritis**, **end-stage** kidney disease, **hearing** loss and sometimes **ocular** disorders
  - **X-linked** dominant (85%), but also AD, **AR**

- Usual age of onset **~5-20 years old**

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Alport Syndrome pathogenesis

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- Defect in synthesis of **type IV collagen** in **GBM**
  - X-linked dominant: Mutation in the gene encoding the α_5-subtype of type IV collagen **(COL4A5)**
  - AR: α_3-subtype of type IV collagen **(COL4A3)**
- Leads to **underproduction** of collagen & **defective assembly** of collagen network

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Alport Syndrome histopathology

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LM: Interstitial (between tubules & glomeruli) **foam cells** due to accumulation of neutral **fats** & **mucopolysaccharides**

IF: Not specific

EM: **Splitting & lamination** of the lamina densa of the GBM giving a **basket-weave** appearance

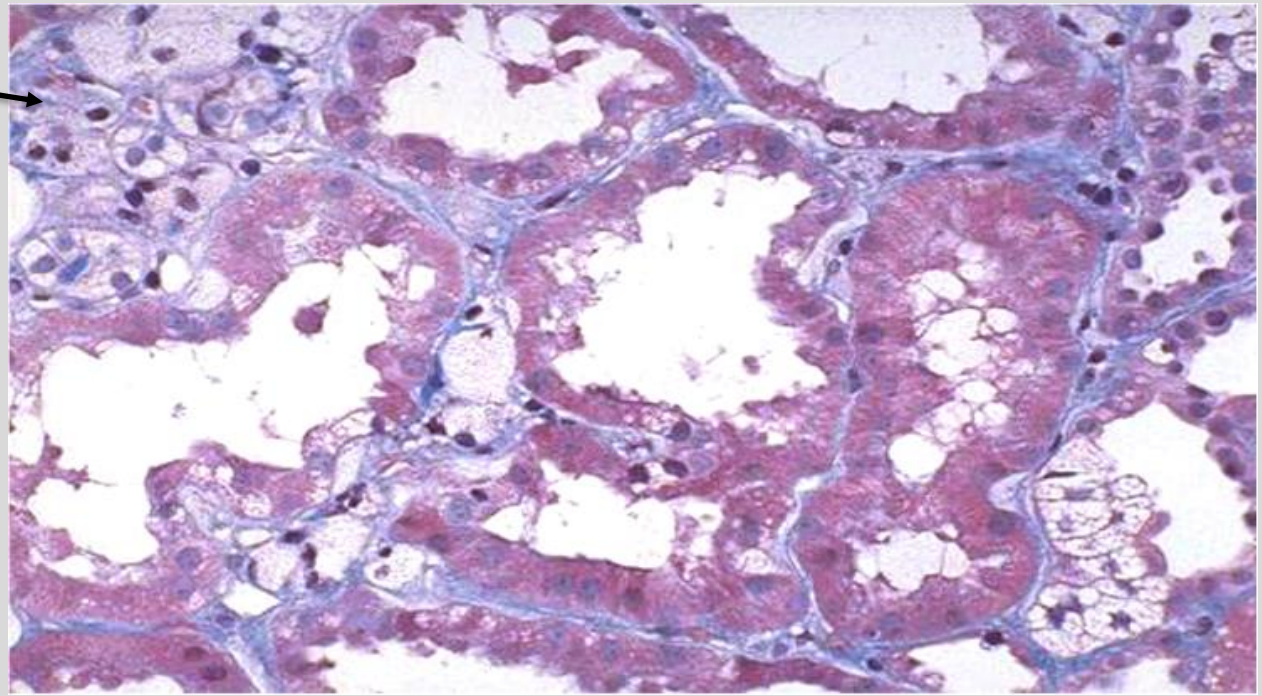
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Alport Syndrome LM

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Foam cells in interstitium

Trichrome stain (nuclei blue/black, cytoplasm magenta, collagen in interstitium blue)



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Alport Syndrome EM

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Basket weave pattern w/ splitting & lamination of the lamina densa



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Alport Syndrome clinical presentation

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- Clinical findings: Sensorineural **hearing** loss, **ocular** abnormalities, **family** history of renal failure

- Example presentation: **7-year-old boy** is brought in by his mother because she notes that he has been having trouble at **school**
  - Recently his teacher moved him to the **front** of the class, and his grades improved. When questioning him, you notice that he sometimes fails to **acknowledge** you
- UA: Many **dysmorphic** RBC's, **RBC casts** & some **protein**
- A **family** history of renal failure is disclosed

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Thin Basement Membrane Disease Overview

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- AKA Benign Familial Hematuria
  - **Hereditary** disorder - **AD**

- Pathogenesis: Anomaly in genes encoding α3- or α4-chains of **type IV collagen**
- Pathology shows extremely **thin GBMs**
  - Also, **hearing** loss, **ocular** abnormalities & **family history** of renal failure are **absent** unlike Alport Syndrome

- Consider (in DDX) in those w/ hematuria &
  - Negative for Berger Disease
  - Negative for Alport Syndrome

- Prognosis: Clinically **find** familial **asymptomatic hematuria,** usually discovered on **routine urinalysis** & histopathologically by **diffuse thinning of GBMs**

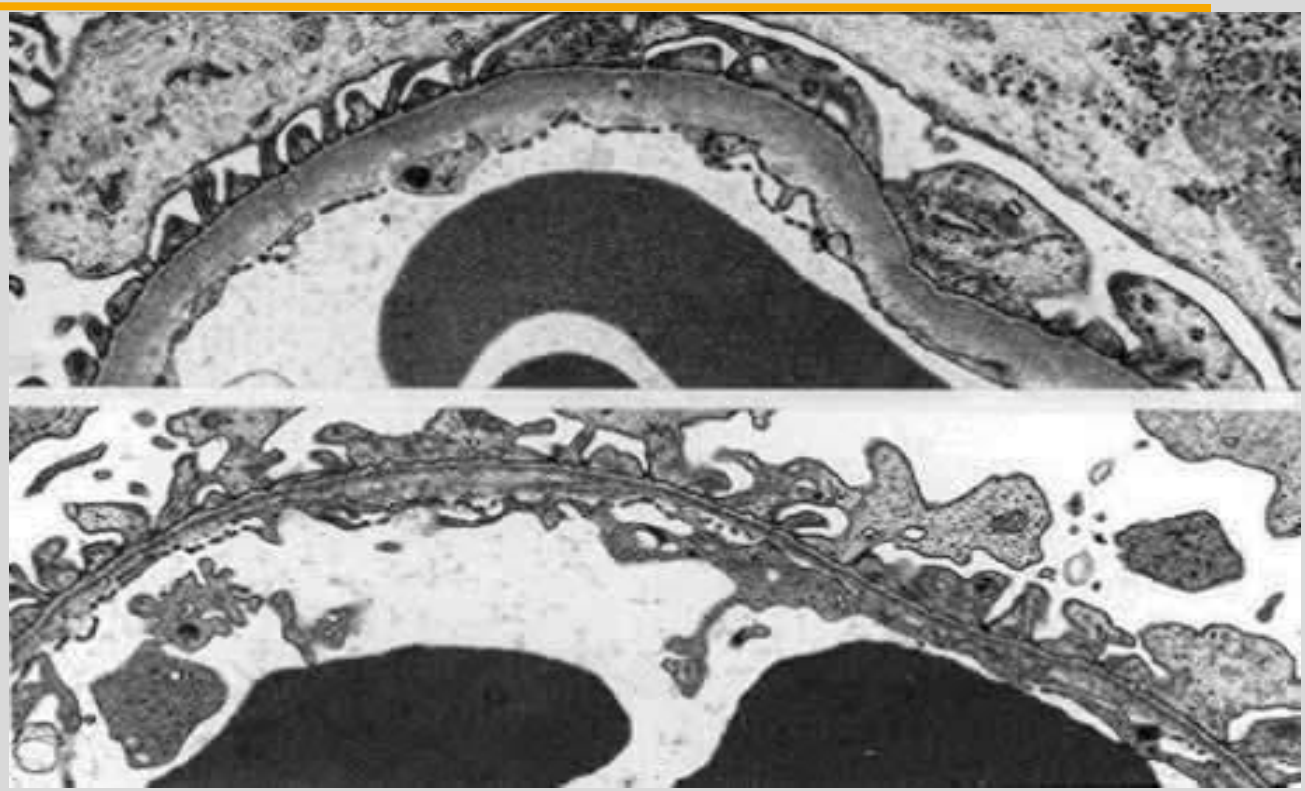
  - Although **mild proteinuria** may be present, renal **function is normal** and **prognosis is excellent**
  - Usually does **not progress** to renal failure

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Thin Basement Membrane Disease EM

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Top **normal**, bottom **thinned** GBM



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Thin Basement Membrane Disease clinical presentation example

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- A **15-year-old boy** has been occasionally passing **red colored urine**
  - **UA shows 1+ hematuria**, no proteinuria and no ketones, glucose, or wbcs
  - His family history reveals that his **mother** & **sister**, who are otherwise **well**, have a **similar condition**

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Thank you,

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